WORKED SOLUTION

STEP I 2011, Question 12

Probability and Statistics • Queues and Ballot Problem

STEP I 2011, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q12

This question has a full guided treatment in *Beyond STEP*, Section 9.10, *Always Able to Give Change*: https://beyondstep.keithmcnulty.org/ch09/ballot.html

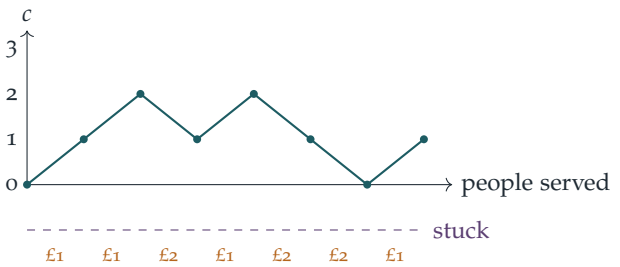
The key idea

Keep track of one number: how many £1 coins I am holding. A £1 customer adds one; a £2 customer takes one away (I keep their £2 coin, which is no use as change). I am stuck exactly when a £2 customer arrives and I am holding no £1 coins. With only a few £2 customers, the danger is all at the front of the queue, so we look at the first few people and work out what each opening leaves us with.

Solution

Only the pattern of coins in the queue matters, not who the people are. Since every arrangement of people is equally likely, the person in any given position is equally likely to be any of the people not yet placed, so we can build the queue from the front: if a people with £1 coins and b with £2 coins remain, the next person has a £1 coin with probability $\frac{a}{a+b}$ and a £2 coin with probability $\frac{b}{a+b}$.

Write c for the number of £1 coins I hold. I can serve everyone if and only if c never needs to drop below 0, that is, at every point in the queue at least as many £1 people as £2 people have been served so far.



The picture shows a successful queue: the path of c never touches the dashed line.

Part (i)

With one £2 customer, the only way to get stuck is if that person is first in the queue, when I hold no change. If anyone else is first, I hold at least one £1 coin by the time the £2 customer arrives. The £2 customer is equally likely to be in any of the $m+1$ positions, so

$$\mathrm{P}(\text{sell to everyone}) = 1 - \frac{1}{m+1} = \frac{m}{m+1}.$$

Part (ii)

Write 1 for a £1 customer and 2 for a £2 customer, and list the openings.

- If the first person is a 2, I am stuck at once.
- If the queue starts $1, 2, 2$, then c goes $1, 0$ and I am stuck at the third person.
- If the queue starts $1, 1$, then $c = 2$ after two people. There are only two £2 customers in total, and c only falls when one of them is served, so c can never fall below $2 - 2 = 0$. Success.
- If the queue starts $1, 2, 1$, then c goes $1, 0, 1$, and only one £2 customer remains, so c never falls below 0. Success.

These cases cover every possibility, so I serve everyone exactly when the queue starts $1, 1$ or $1, 2, 1$. Building the queue from the front,

$$\mathrm{P}(1,1) = \frac{m}{m+2} \cdot \frac{m-1}{m+1}, \quad \mathrm{P}(1,2,1) = \frac{m}{m+2} \cdot \frac{2}{m+1} \cdot \frac{m-1}{m} = \frac{2(m-1)}{(m+2)(m+1)}.$$

These events are disjoint, so

$$\mathrm{P}(\text{sell to everyone}) = \frac{m(m-1) + 2(m-1)}{(m+2)(m+1)} = \frac{(m-1)(m+2)}{(m+2)(m+1)} = \frac{m-1}{m+1},$$

as required. (We need $m \geq 2$ so that $m - 1 \geq 1$ £1 people remain in the last factor.)

Part (iii)

Now there are three £2 customers. Again look at the first three people.

- Starts with 2, or with $1,2,2$: stuck, exactly as in (ii).
- Starts $1,1,1$: then $c = 3$, and the three £2 customers can take c down to 0 at worst. Success.
- Starts $1,1,2$ or $1,2,1$: in both cases $c = 1$ after three people, and what remains is $m-2$ people with £1 coins and 2 with £2 coins, in random order.

Ask yourself In the last case, when does the rest of the queue go wrong?

Starting the rest of the queue with $c = 1$ is like having one free £1 customer at the front of a queue with two £2 customers, which is part (ii) in disguise. With $c = 1$ and only two £2 customers left, I get stuck exactly when both of the next two people are £2 customers: c goes 0, then I cannot serve. If either of the next two people has a £1 coin, then c never falls below 0 (check $1,\ldots$ and $2,1,\ldots$ as in (ii)). Among the m people remaining after the first three,

$$\mathrm{P}(\text{next two are both 2}) = \frac{2}{m}\cdot\frac{1}{m-1},$$
$$\text{so}\quad \mathrm{P}(\text{rest succeeds}) = 1 - \frac{2}{m(m-1)} = \frac{(m-2)(m+1)}{m(m-1)},$$

using $m^2 - m - 2 = (m-2)(m+1)$.

Write $D = (m+3)(m+2)(m+1)$. The probabilities of the useful openings are

$$\mathrm{P}(1,1,1) = \frac{m(m-1)(m-2)}{D},$$
$$\mathrm{P}(1,1,2) = \frac{m\cdot(m-1)\cdot 3}{D}, \qquad \mathrm{P}(1,2,1) = \frac{m\cdot 3\cdot(m-1)}{D}.$$

Since the openings are disjoint and the rest of the queue is again in random order whatever the opening,

$$\begin{aligned}\mathrm{P}(\text{success}) &= \frac{m(m-1)(m-2)}{D} + \frac{6m(m-1)}{D}\cdot\frac{(m-2)(m+1)}{m(m-1)}\\ &= \frac{(m-2)\big[m(m-1)+6(m+1)\big]}{D} = \frac{(m-2)(m^2+5m+6)}{D}\\ &= \frac{(m-2)(m+2)(m+3)}{(m+3)(m+2)(m+1)} = \frac{m-2}{m+1},\end{aligned}$$

as required. (Here $m \geq 3$ ensures $m-1 \geq 2$, so the conditional probability makes sense.)

Checks

- Smallest cases: with $m = n$ the formulas give $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ for $n = 1,2,3$. For $m = n = 2$ the six patterns of 1s and 2s are equally likely, and only 1122 and 1212 succeed: $\frac{2}{6} = \frac{1}{3}$. ✓

- Large m: all three probabilities tend to 1, as they should when £1 coins are plentiful.
- By complete enumeration of queue orders, for $m = 5$ I get $\frac{5}{6}$, $\frac{2}{3}$, $\frac{1}{2}$ for $n = 1, 2, 3$, matching $\frac{m}{m+1}$, $\frac{m-1}{m+1}$, $\frac{m-2}{m+1}$; the formulas were also checked exactly for every m up to 14.

Common slips

- Forgetting that the £2 coins I collect are useless as change: only £1 coins count.
- Treating the people as independent draws with fixed probability $\frac{m}{m+n}$. The queue is drawn without replacement, so the probabilities change as people are placed.
- In (iii), assuming that $c = 1$ after three people is always safe, and missing the failure $1, 1, 2, 2, 2$ or $1, 2, 1, 2, 2$.

Beyond STEP

The pattern is $\frac{m+1-n}{m+1}$ for every $n \leq m$. This is the *ballot problem*, and the neatest proof uses the *reflection principle*: a path that touches the stuck level can be reflected to give a path ending somewhere else, which turns the count of bad queues into a single binomial coefficient. When $m = n$ the number of good queues is a Catalan number, which turns up all over combinatorics. For random walks and the reflection principle, see Grimmett and Welsh, *Probability: An Introduction*.

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